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The size of the problem is n (board of size $2^n \times 2^n$). We can solve the problem by recursion. The base case is n = 1. It is a 2×2 corner-covered board. We can cover it with one triominoe and solve the problem. In the recursion step, divide the corner-covered board of size $2^n \times 2^n$ into 4 sub-boards, each of size $2^{n-1} \times 2^{n-1}$, by drawing horizontal and vertical lines through the centre of the board. Place atriominoe at the center of the entire board so as to not cover the corner-covered sub-board, as shown in the left-most board of Figure 8.11. Now, we have four corner-covered boards, each of size $2^{n-1} \times 2^{n-1}$.

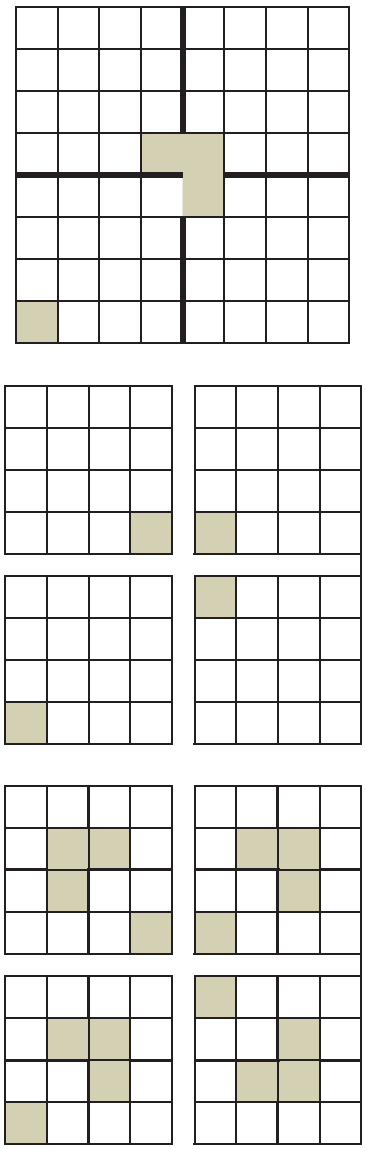
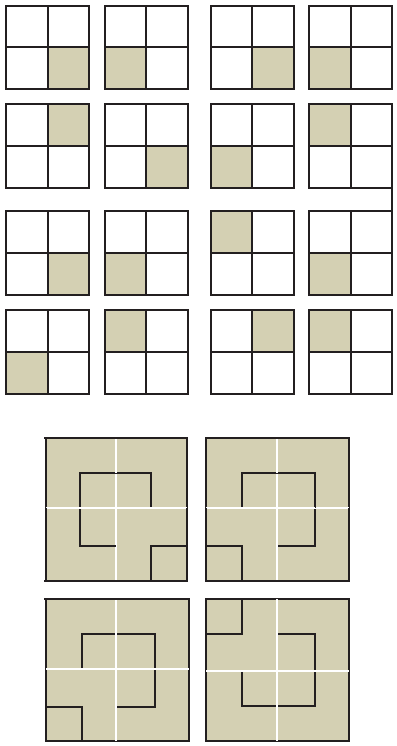


Figure 8.11: Recursive process of covering a corner-covered board of size 2 x 2^3



We have 4 sub-problems whose size is strictly smaller than the size of the given problem. We can solve each of the sub-problems recursively.

tile corner_covered board of size n

if n = 1 -- base case

cover the 3 squares with one triominoe

else -- recursion step

divide board into 4 sub_boards of size n-1

place a triominoe at centre of board ,

leaving out the corner_covered sub -board

tile each sub_board of size n-1

The resulting recursive process for covering a 2^3 x 2^3 corner-covered board is illus*trated in Figure 8.11.*